Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **object-oriented** software in **Python**, **Java**, and **C++**, with experience in REST APIs, relational SQL, automated **unit testing**, debugging, and Agile delivery across web, desktop, and simulation systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **relational database** schemas with row-level data isolation for a multi-tenant platform serving 500+ users, cutting dashboard latency from 1.2s to 350ms through SQL query and indexing optimization.
- Built fault-tolerant Python job orchestration with **explicit state transitions** and failure isolation, tripling throughput across 1K+ jobs, and debugged integration pipelines with **validation checks** sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed a **React** web interface and Flask **REST APIs** for upload, grading, and history workflows, integrating multimodal LLMs and OCR to serve 150+ instructors and 10K+ daily requests on AWS.
- Structured a **modular MVC** Flask backend with multithreaded batch processing and secure PDF, DOCX, and ZIP ingestion, handling 1K+ files per batch and cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python and PyTorch pipelines processing 100K+ autonomous-driving trajectory logs into features for a Decision Transformer **motion planner**, improving long-horizon trajectory planning 18% in closed-loop simulation testing.
- Designed a modular spatial-temporal encoder fusing HD-map geometry with ego-vehicle trajectories via multi-head attention, benchmarking against three baseline encoders in **closed-loop simulation** and improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **object-oriented** modules using Singleton and Factory patterns for cross-platform data-sharing software, validated through 20+ automated **unit and integration test** suites with PyTest and Mockito across Windows, Linux, and macOS.
- **Troubleshot** and optimized microservice API integrations with pagination, field filtering, and caching, improving response times 30% while running nightly Jenkins builds that installed executables on test VMs in **Agile** sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a React and **Java Spring Boot** resource-management platform with RESTful microservices, real-time WebSocket updates, and Azure SQL persistence, synchronizing 2M+ records across locations for multi-site operations.
- Automated Jenkins CI/CD with unit, integration, and **end-to-end tests** in Jest and Selenium before Docker packaging and staged releases, while analytics-driven fixes reduced resource wastage 42% and raised lead conversions 15%.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Bash
Software Engineering: Object-Oriented Design, Design Patterns, Data Structures, Algorithms, Debugging, Code Reviews
Testing & Quality: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Jest, Selenium, CI/CD, Jenkins
Web & Data: REST APIs, Spring Boot, Flask, FastAPI, React, PostgreSQL, SQL Server, MySQL, Azure SQL
Tools & Practices: Git, GitHub Actions, Docker, Linux, AWS, Azure, TCP/IP, Agile/Scrum, Documentation

Projects

Lab Automation Control Simulator | *C++, Python, State Machines, TCP/IP, Unit Testing, Git*

- Designed object-oriented **C++** control software for a simulated sample-handling robot, modeling motion, run-state, and fault-recovery logic behind a hardware abstraction layer with TCP/IP device messaging and automated **unit tests**.

Traceable Lab Results Service | *Java, Spring Boot, REST, PostgreSQL, React, Mockito*

- Designed a **REST** web service and React interface for lab result review with PostgreSQL storage, role-based access, immutable audit trails, and requirement-traced automated tests supporting **verification and validation** documentation.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker*

- Extended a cross-platform **desktop** database workspace in Java and TypeScript supporting 10+ engines across Windows, macOS, and Linux, with schema synchronization and **SQL** generation reaching 82% accuracy.